

LatticeMico32 Tri-Speed Ethernet Media Access Controller

The LatticeMico32 Tri-Speed Ethernet media access controller (MAC) peripheral is an open-source IP core that includes all logic blocks necessary to enable LatticeMico32 to interface seamlessly to an Ethernet PHY. The peripheral core includes an ispLEVER-generated Tri-Speed MAC IP core, 8-kilobyte receive and transmit FIFO buffers, 32-bit receive and transmit statistics counter registers, system control and status registers, and an OpenCores 32-bit WISHBONE-compatible system-on-a-chip (SoC) slave interconnect that is used to interface to these blocks. The LatticeMico32 Tri-Speed MAC peripheral core, in conjunction with LatticeMico32 software (TCP/IP stack), provides the means for the LatticeMico32 software to interface easily to a fast Ethernet network.

Features

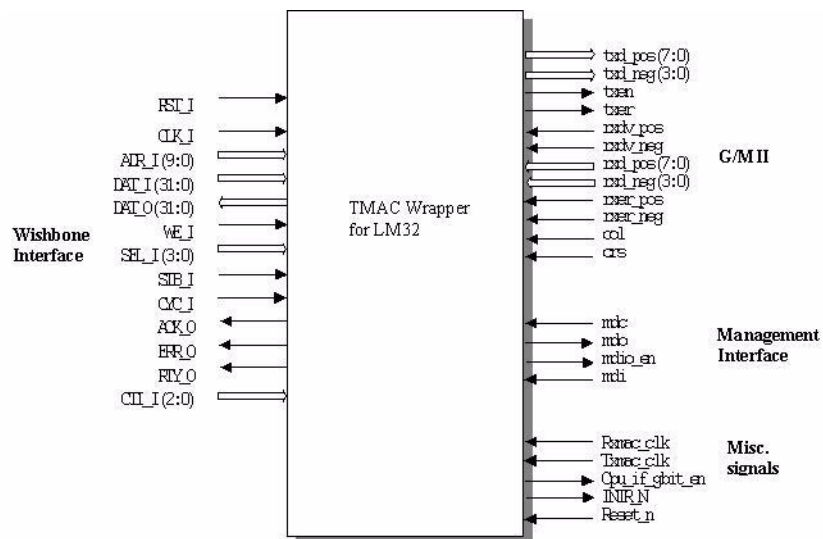
The LatticeMico32 Tri-Speed MAC peripheral includes the following features:

- ◆ Compliant to IEEE 802.3z standard
- ◆ 32-bit WISHBONE-compatible slave interface
- ◆ WISHBONE-accessible 8-kilobyte transmit and receive FIFO buffers
- ◆ WISHBONE-accessible 32-bit transmit and receive statistics counters
- ◆ WISHBONE-accessible internal Tri-Speed MAC core registers
- ◆ WISHBONE-accessible system control and status registers
- ◆ Supports the following:
 - ◆ Multicast address filtering
 - ◆ Programmable inter-packet gap (IPG)
 - ◆ Full- and half-duplex operation in 10/100 mode

- ◆ Full-duplex control using pause frames
- ◆ VLAN tagged frames
- ◆ Automatic re-transmission on collision
- ◆ Automatic padding of short frames
- ◆ Multicast and broadcast frames
- ◆ Optional FCS transmission and reception
- ◆ Optional MII management interface module
- ◆ Jumbo frames up to 8192 bytes

Figure 1 shows all the interfaces and signal ports in the LatticeMico32 Tri-Speed MAC.

Figure 1: LatticeMico32 Tri-Speed MAC

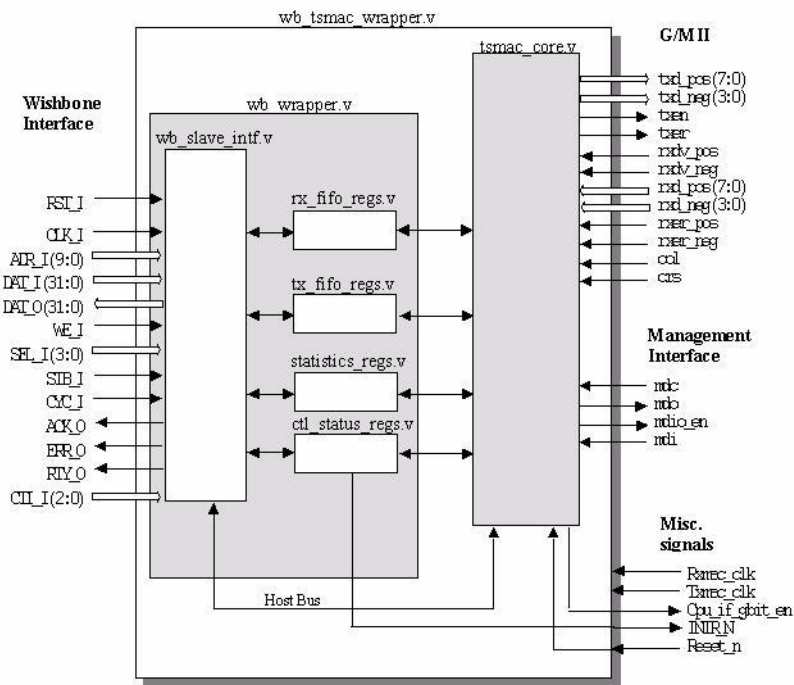


Functional Description

Block Diagram

Figure 2 shows a block diagram of the wb_tsmac_wrapper with its different levels of hierarchy. Figure 3 shows a more functional representation of the basic architecture of the wrapper design.

Figure 2: Block Diagram of the LatticeMico32 Tri-Speed MAC



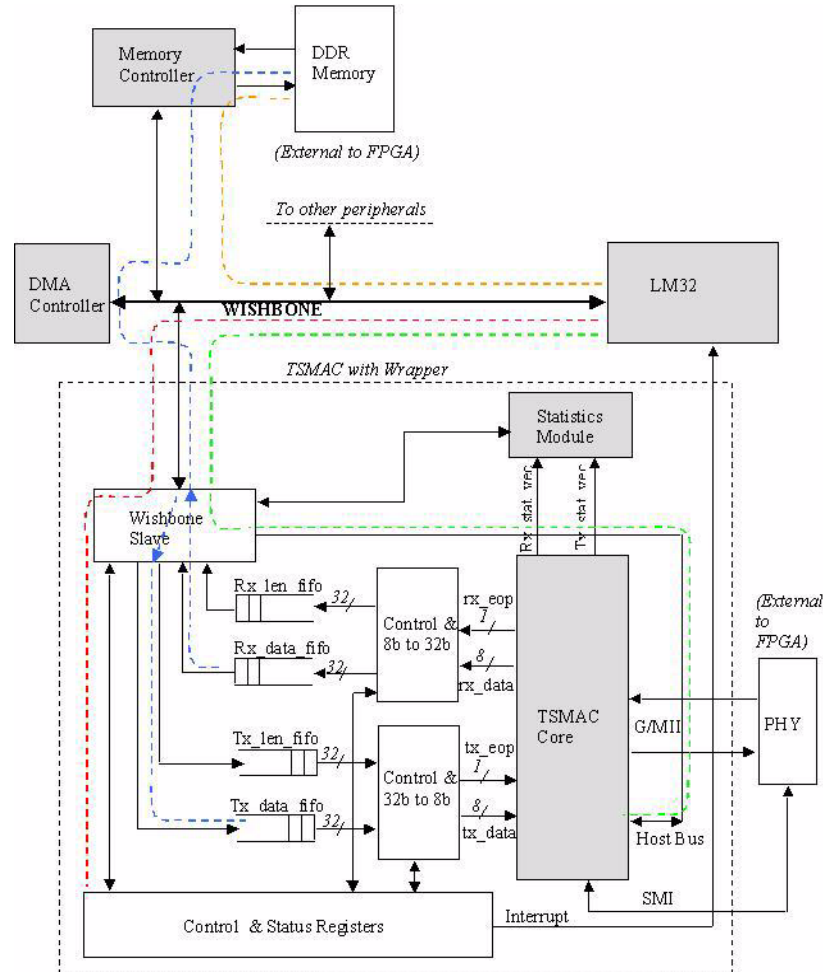
Basic Architecture

The basic architecture of the LatticeMico32 Tri-Speed MAC is shown in Figure 3.

- ◆ The blue dashed line shows the data flow to and from the G/MII and external memory.
- ◆ The orange dashed line shows the data flow to and from the LatticeMico32 microprocessor and the external memory. This flow is defined as the LatticeMico32 data path.
- ◆ The green dashed line shows the control and status data flow to and from the Tri-Speed MAC and the LatticeMico32 microprocessor. This flow is defined as the Tri-Speed MAC control path.
- ◆ The red dashed line shows the control and status data flow to and from the registers and the LatticeMico32 microprocessor. This flow is defined as the register control path.

- ◆ The Rx data path is defined as the Ethernet data from the G/MII to memory, and the Tx data path is defined as the Ethernet data from the memory to G/MII.

Figure 3: Architecture of the LatticeMico32 Tri-Speed MAC



Basic Receive Flow

The basic receive (Rx) flow is as follows:

1. A receive packet is received and processed by the Tri-Speed MAC.
2. The Tri-Speed MAC checks to see if there is space available in the Rx_data FIFO. If there is, it begins writing the packet into the Rx_data FIFO. If the Rx_data FIFO is almost full, an interrupt is generated to the LatticeMico32 microprocessor, and the Tri-Speed MAC may assert the ignore_next_pkt bit in the SYS_CTL register (the receive Tri-Speed MAC drops packets when this signal is asserted), as well as signal the transmit Tri-Speed MAC to transmit pause frames to PHY by setting the tx_sndpausreq bit in the SYS_CTL register.
3. The Control-and-8b-to-32b block converts the 8-bit frame data to 32 bits and conditions the write enable to the 32-bit FIFO to write the data into the

data FIFO appropriately. The length of the frame is calculated from the write enable and EOP.

4. After the frame is completely written into the Rx_data FIFO, the Control-and-8b-to-32b block writes the length of the frame into the Rx_len FIFO and sends an interrupt to the LatticeMico32 microprocessor, indicating that a receive frame is ready. A Rx_frame counter register is also incremented.
5. Once the LatticeMico32 microprocessor services the interrupt, it must read the Rx_frame counter register to make sure a receive frame is available (if interrupts are not desired, the LatticeMico32 microprocessor can just poll the Rx_frame counter to see if there are any available receive frames). The LatticeMico32 microprocessor then reads the Rx_len FIFO to determine how many bytes need to be read from the Rx_data FIFO. Once the length has been read, the LatticeMico32 microprocessor can set up a DMA transfer from the Rx_data FIFO to the DDR memory, or read the Rx_data FIFO and write to the memory itself. Once the Rx_len FIFO is read, the Rx_frame counter is decremented.

Basic Transmit Flow

The basic transmit (Tx) flow is as follows:

1. A transmit packet is ready to be transmitted by the LatticeMico32 microprocessor.
2. The LatticeMico32 microprocessor first checks to see if there is space available in the Tx_data FIFO. If there is, it begins writing the packet into the Tx_data FIFO (this may be a DMA transfer).

Note

The LatticeMico32 microprocessor checks the status of the Tx_data FIFO by reading the Tx_fifo_status register or the Tx_frame counter register. If the Tx_data FIFO is almost full or almost empty, an interrupt is generated to the LatticeMico32 microprocessor. In addition, the Tx_fifo_status register indicates the almost full or empty status.

3. After the frame is completely written into the Tx_data FIFO, the LatticeMico32 microprocessor writes the length of the frame into the Tx_len FIFO. When the length is written into the Tx_len FIFO, a Tx_frame counter register is incremented. The Control-and-32b-to-8b block polls the Tx_frame counter register to see if a frame is available to be transmitted.
4. If a frame is available to be transmitted, the Control-and-32b-to-8b block reads the length of the frame from the Tx_len FIFO and indicates to the Tri-Speed MAC that the Tx FIFO is not empty by de-asserting the Tx_fifoempty signal and by asserting the tx_fifoavail signal.
5. When the Tri-Speed MAC asserts the tx_macread signal, the Control-and-32b-to-8b block converts the 32-bit frame data to 8 bits and conditions the read enable to the 32-bit FIFO to read the data out of the FIFO appropriately. At the end of the frame, the EOP is asserted in the correct location by using the frame length information from the Tx_len FIFO. The Tx_frame counter register is also decremented, and an interrupt is

generated to the LatticeMico32 microprocessor to indicate that the frame was transmitted.

Configuration

The following sections describe the graphical user interface (UI) parameters, the hardware description language (HDL) parameters, and the I/O ports that you can use to configure and operate the LatticeMico32 Tri-Speed MAC.

UI Parameters

Table 1 shows the UI parameters available for configuring the LatticeMico32 Tri-Speed MAC through the Mico System Builder (MSB) interface.

Table 1: LatticeMico32 Tri-Speed MAC UI Parameters

Dialog Box Option	Description	Allowable Values	Default Value
Instance Name	Specifies the name of the Tri-Speed MAC instance.	Alphanumeric and underscores	
Base Address	Specifies the base address for the Tri-Speed MAC instance. The minimum byte alignment is 0X80.	0X80000000 – 0XFFFFFFF8 If other components are included in the platform, the range of allowable values will vary.	0X80000000
Include Statistics Regs	Sets a parameter to include the statistics_regs.v module in the wb_wrapper.v module. These statistics counter registers are described in the <i>IPUG51 - Tri-speed Ethernet Media Access Controller User's Guide</i> in the sections on the transmit and receive statistics vectors.	0 1	1
Use MDIO Interface	Includes the MDIO interface in the Tri-Speed MAC IP core generated by IPexpress. The MDIO interface is a standard IEEE 802.3 serial management interface consisting of an mdc signal (low-speed clock), an mdi (data_in) signal, an mdo (data_out) signal, and an mdio_en (bidirectional enable) signal. This interface allows the Tri-Speed MAC IP core to interface to other Ethernet devices, like a PHY device, and configure and monitor its registers. Access to the MDIO interface on the TSMAC IP core client side is through the host bus interface described in the <i>IPUG51 - Tri-speed Ethernet Media Access Controller User's Guide</i> . When the TSMAC wrapper is used with the WISHBONE interface, access to the MDIO interface is through the WISHBONE bus.	0 1	1

HDL Parameters

Table 2 lists the parameters that appear in the HDL.

Table 2: LatticeMico32 Tri-Speed MAC HDL Parameters

Parameter Name	Description	Allowable Values
STAT_REGS	Sets a parameter to include the statistics_regs.v module in the wb_wrapper.v module. These statistics counter registers are described in the <i>IPUG51 - Tri-speed Ethernet Media Access Controller User's Guide</i> in the sections on the transmit and receive statistics vectors.	0 1
MIIM_MODULE	Includes the MDIO interface in the Tri-Speed MAC IP core generated by IPexpress. The MDIO interface is a standard IEEE 802.3 serial management interface consisting of an mdc signal (low-speed clock), an mdi (data_in) signal, an mdo (data_out) signal, and an mdio_en (bidirectional enable) signal. This interface allows the Tri-Speed MAC IP core to interface to other Ethernet devices, like a PHY device, and configure and monitor its registers. Access to the MDIO interface on the TSMAC IP core client side is through the host bus interface described in the <i>IPUG51 - Tri-speed Ethernet Media Access Controller User's Guide</i> . When the TSMAC wrapper is used with the WISHBONE interface, access to the MDIO interface is through the WISHBONE bus.	0 1

I/O Ports

Table 3 describes the input and output ports of the LatticeMico32 Tri-Speed MAC.

Table 3: LatticeMico32 Tri-Speed MAC Signals

Signal	Type	Active State	Description
rxmac_clk	Input	N/A	The receive Tri-Speed MAC application interface clock is used by the Tri-Speed MAC and internal logic. All outputs driven by the receive Tri-Speed MAC to the internal interfaces (receive FIFO to MAC and so forth) are synchronous to this clock. This clock's frequency is 125, 12.5, or 1.25 MHz, depending on whether the mode is 1G, 100, or 10, respectively. Note: This clock can be viewed as a byte clock, because all receive Tri-Speed MAC bytes are aligned with this clock. This clock is derived from the system G/MII rx_clk.
txmac_clk	Output	N/A	The transmit Tri-Speed MAC application interface clock is used by the Tri-Speed MAC and internal logic. All internal inputs to the transmit Tri-Speed MAC on the internal interface (transmit FIFO to MAC and so forth) should be synchronous to this clock. This clock's frequency is 125, 12.5, or 1.25 MHz, depending on whether the mode is 1G, 100, or 10, respectively. Note: This clock can be viewed as a byte clock, because all transmit Tri-Speed MAC bytes should be aligned with this clock. This clock is derived from the system sys_clk or tx_clk (1G or 10/100, respectively).
cpu_if_gbit_en	Output	High	The CPU interface gigabit -mode-enabled signal, when high, is an indication from the Tri-Speed MAC that the 1-gigabit mode is enabled. It reflects the state of bit 0 of the Tri-Speed MAC mode register.
mdc	Input	N/A	The management data clock is used only when the management interface module is implemented.
INTR_N	Output	Low	The interrupt to the LatticeMico32 microprocessor (see "Register Definitions" on page 12 and "Interrupt Hierarchy" on page 36).
reset_n	Input	Low	The reset is an active low signal that resets the internal registers and internal logic in the Tri-Speed MAC IP core. When activated, the I/O signals are driven to their inactive levels.

Table 3: LatticeMico32 Tri-Speed MAC Signals

Signal	Type	Active State	Description
WISHBONE Interface			
RST_I	Input	High	The reset is an active high WISHBONE signal that resets the internal WISHBONE-related registers and internal logic. When activated, the I/O signals are driven to their inactive levels. You can invert this signal inside the wrapper and tie it to reset_n, if you wish.
CLK_I	Input	N/A	The WISHBONE clock input [CLK_I] coordinates all activities for the internal logic within the WISHBONE interconnect. All WISHBONE output signals are registered at the rising edge of [CLK_I]. All WISHBONE input signals are stable before the rising edge of [CLK_I].
ADR_I[11:2]	Input	N/A	The address input array [ADR_I()] is used to pass a binary address. The higher array boundary is specific to the address width of this wrapper, and the lower array boundary is determined by the data port size. In this wrapper, there is a 32-bit data port with byte granularity.
DAT_I[31:0]	Input	N/A	The data bus input array [DAT_I()] is used to pass binary data. The array boundaries are determined by the port size. The port size of this wrapper is 32 bits (for example, [DAT_I(31..0)]). See also the [DAT_O()] and [SEL_O()] signal descriptions.
DAT_O[31:0]	Output	N/A	The data output array [DAT_O()] is used to pass binary data. The array boundaries are determined by the port size. The port size of this wrapper is 32 bits (for example, [DAT_I(31..0)]). See also the [DAT_I()] and [SEL_O()] signal descriptions.
WE_I	Input	High	The write enable input. [WE_I] indicates whether the current local bus cycle is a read or write cycle. The signal is negated during read cycles and asserted during write cycles.
SEL_I(3:0)	Input	High	The select input array [SEL_I()] indicates where valid data is placed on the [DAT_I()] signal array during write cycles and where it should be present on the [DAT_O()] signal array during read cycles. The array boundaries are determined by the granularity of a port. In this wrapper, 8-bit granularity is used on a 32-bit port, so there is an array of four select signals with boundaries of [SEL_I(3..0)]. Each individual select signal correlates to one of four active bytes on the 32-bit data port. See also the [STB_I] signal descriptions.

Table 3: LatticeMico32 Tri-Speed MAC Signals

Signal	Type	Active State	Description
STB_I	Input	High	The strobe input [STB_I], when asserted, indicates that the slave is selected. A slave responds to other WISHBONE signals only when this [STB_I] is asserted, except for the [RST_I] signal, which should always be responded to. The slave asserts either the [ACK_O], [ERR_O], or [RTY_O] signals in response to every assertion of the [STB_I] signal.
CYC_I	Input	High	The cycle input [CYC_I], when asserted, indicates that a valid bus cycle is in progress. The signal is asserted for the duration of all bus cycles. For example, during a block transfer cycle, multiple data transfers can occur. The [CYC_I] signal is asserted during the first data transfer and remains asserted until the last data transfer.
ACK_O	Output	High	The acknowledge output [ACK_O], when asserted, indicates the termination of a normal bus cycle. See also the [ERR_O] and [RTY_O] signal descriptions.
ERR_O	Output	High	The error output [ERR_O] indicates an abnormal cycle termination. The source of the error and the response generated by the master is defined by the IP core supplier. See also the [ACK_O] and [RTY_O] signal descriptions.
RTY_O	Output	High	The retry output [RTY_O] indicates that the interface is not ready to accept or send data and that the cycle should be retried. In this wrapper, the cycle should be retried after 20 CLK_I ticks. See also the [ERR_O] and [RTY_O] signal descriptions.
CTI_I[2:0]	Input	N/Z	The cycle type identifier [CTI_IO()] address tag provides additional information about the current cycle. The master sends this information to the slave. The slave can use this information to prepare the response for the next cycle. CTI_O(2:0) description: ◆ 000 Classic cycle ◆ 001 Constant address burst cycle ◆ 010 Incrementing burst cycle ◆ 011 Reserved ◆ 100 Reserved ◆ 101 Reserved ◆ 110 Reserved ◆ 111 End-of-burst Note: Only the classic cycle is supported in this wrapper.

Table 3: LatticeMico32 Tri-Speed MAC Signals

Signal	Type	Active State	Description
Management Interface Signals			
mdi	Input	High	The management data input is used to transfer information from the PHY to the management module.
mdo	Output	High	The management data output is used to transmit information from the management module to the PHY.
mdio_en	Output	High	The management data out enable is asserted whenever mdo is valid. It can be used to implement a bidirectional signal for mdi and mdo.
G/MII Signals			
txd_pos[7:0] txd_neg[3:0]	Output	High	txd_pos[7:4] is the transmit data sent to the PHY chip – high nibble. In gigabit mode, these bits are used as the GMII txd[7:4] bits after they are pipelined outside the wrapper through some I/O flip-flops clocked at 125 MHz (txmac_clk). These bits are not used in 10/100 mode. See the <i>IPUG51 - Tri-speed Ethernet Media Access Controller User's Guide</i>. txd_pos[3:0], txd_neg[3:0] are the transmit data sent to the PHY chip – low nibble. In both gigabit mode and 10/100 mode, both the txd_pos[3:0] and txd_neg[3:0] bits are used to generate the G/MII txd[3:0] bits after they are multiplexed outside the wrapper through some I/O DDR cells. In 1-gigabit mode, txd_pos[3:0] and txd_neg[3:0] always have the same value, but in 10/100 mode, the txd_pos[3:0] has the high nibble of the byte transmitted, and txd_neg[3:0] has the low nibble. In 1-gigabit mode, the txmac_clk rate is 125 MHz, and in the 10/100 mode, the clock rate is 1.25 MHz and 12.5 MHz, respectively. See the <i>IPUG51 - Tri-speed Ethernet Media Access Controller User's Guide</i>.
txen	Output	High	The transmit enable is asserted by the Tri-Speed MAC to indicate that the txd bus contains a valid frame.
txer	Output	High	The transmit error is asserted when the Tri-Speed MAC generates a coding error on the byte currently being transferred.
rxdv_pos rxdv_neg	Input	High	The receive data valid indicates that the data on the rxd bus is valid. This signal is pipelined before entering the wrapper with the rx_clk. The _pos signals are sampled on the rising edge of rxmac_clk, and the _neg signals are sampled on the falling edge of rxmac_clk before being presented to the Tri-Speed MAC wrapper. See the <i>IPUG51 - Tri-speed Ethernet Media Access Controller User's Guide</i> .

Table 3: LatticeMico32 Tri-Speed MAC Signals

Signal	Type	Active State	Description
rxd_pos[7:0] rxn_neg[3:0]	Input	N/A	Receive data bus. Data is driven by the PHY on these lines and is valid whenever rxdv is asserted. These signals are pipelined before entering the wrapper with the rxmac_clk. The _pos signals are sampled on the rising edge of rxmac_clk, and the _neg signals are sampled on the falling edge of rxmac_clk before being presented to the Tri-Speed MAC core. See the <i>IPUG51 - Tri-speed Ethernet Media Access Controller User's Guide</i> .
rxer_pos rxer_neg	Input	High	The receive data error signal is asserted by the external PHY device when it detects an error during frame reception. This signal is pipelined before entering the wrapper with the rxmac_clk. The _pos signals are sampled on the rising edge of rxmac_clk, and the _neg signals are sampled on the falling edge of rxmac_clk before being presented to the Tri-Speed MAC core. See the <i>IPUG51 - Tri-speed Ethernet Media Access Controller User's Guide</i> .
col	Input	High	The collision is an active-high signal that indicates that a collision occurred during transmission. This signal is valid for half-duplex operations in fast Ethernet (10/100) mode only. Otherwise, it is ignored.
crs	Input	High	The carrier sense, when at logic high, indicates that the network has activity. Otherwise, it indicates that the network is idle. This signal is valid for half-duplex operations in fast Ethernet (10/100) mode only.

Register Definitions

The LatticeMico32 Tri-Speed MAC has registers in four distinct memory mapped spaces:

- ◆ Tx_data, Rx_data, Tx_len, and Rx_len FIFO register
- ◆ Control and status registers
- ◆ Tri-Speed MAC IP core registers (registers inside the MAC IP core)
- ◆ Statistics counter registers

The Tx_data, Rx_data, Tx_len, and Rx_len FIFO registers are 32-bit data and length FIFOs, with fixed addresses. The LatticeMico32 microprocessor firmware accesses these FIFOs through the transmit and receive data paths.

The control and status registers are 32-bit registers used to control the reception and transmission of Ethernet frames from and to the DDR memory and to and from the FIFOs. The LatticeMico32 microprocessor firmware accesses these control and status registers through the register control path.

The Tri-Speed MAC registers enable, configure, and monitor the Tri-Speed MAC. Since each Tri-Speed MAC register consists of two 8-bit registers (byte_0 and byte_1) on word boundaries, a single read or write to these registers through the 32-bit WISHBONE (Tri-Speed MAC control path) is an atomic operation that consists of two 8-bit register accesses, handled by a separate finite state machine. Two registers are used to access the Tri-Speed MAC registers:

- ◆ The MAC_REG_ADDR_RW register contains the read or write address of the register to be accessed (the address written to this register is on word boundaries), and a read/write control bit that indicates a read (low) or write (high) operation.
- ◆ The MAC_REG_DATA register contains the data read back or the data to write to the address specified in the MAC_REG_ADDR register.

A Tri-Speed MAC register read consists of the following:

- ◆ Writing the MAC_REG_ADDR_RW register with the appropriate address and with the read/write bit set to 0. Writing the MAC_REG_ADDR_RW register kicks off the atomic operation, and if an atomic operation is already in progress, the WISHBONE RTY_O (retry) signal is asserted to indicate that the Tri-Speed MAC host bus is busy and that the access will not be performed.
- ◆ Reading the MAC_REG_DATA register (one word read back). If the data from the atomic read is not yet valid, the WISHBONE RTY_O (retry) signal is asserted to indicate that the Tri-Speed MAC host bus is busy and that the access will not be performed.

A Tri-Speed MAC register write consists of: the following:

- ◆ Writing the MAC_REG_DATA register with the two bytes to be written. If an atomic operation is already in progress, the WISHBONE RTY_O (retry) signal is asserted to indicate that the Tri-Speed MAC host bus is busy and that the data just written will not be used.
- ◆ Writing the MAC_REG_ADDR_RW register with the appropriate address and with the read/write bit set to 1. Writing the MAC_REG_ADDR_RW register kicks off the atomic operation, and if an atomic operation is already in progress, the WISHBONE RTY_O (retry) signal is asserted to indicate that the Tri-Speed MAC host bus is busy and that the access will not be performed.

Note

The additional processor clock cycles involved in writing to the Tri-Speed MAC registers should not be an issue since these registers are primarily intended to be configured once and then ignored.

The statistics counter registers are 32-bit registers used to keep received and transmitted packet statistics. The LatticeMico32 microprocessor firmware accesses these control and status registers through the register control path. The statistics counter registers reside in a separate module within the Tri-Speed MAC wrapper, and you can pull them out, if you wish, with the Include Statistics Regs option in the graphical user interface.

Data organization of the 32-bit port conforms to the little endian ordering shown in Table 4.

Table 4: 32-Bit Data Bus with 8-Bit (Byte) Granularity

Address Range		Active Portion of Data Bus			
ADR_I	DAT_I	DAT_I	DAT_I	DAT_I	DAT_I
ADR_O	DAT_O	DAT_O	DAT_O	DAT_O	DAT_O
(31..2)	(31..24)	(23..16)	(15..08)	(07..00)	
Active	SEL_I(3)	SEL_I(2)	SEL_I(1)	SEL_I(0)	
Select line	SEL_O(3)	SEL_O(2)	SEL_O(1)	SEL_O(0)	
Byte Ordering	Little endian	Byte(3)	Byte(2)	Byte(1)	Byte(0)
		Byte(7)	Byte(6)	Byte(5)	Byte(4)

Table 5 describes the Tri-Speed MAC registers.

Table 5: LatticeMico32 Tri-Speed MAC Register Descriptions

Address [11:0] (Note: only [11:2] is decoded)	Register Name	Description	Size	Type
FIFO Registers				
0x000	RX_LEN_FIFO	Receive packet-length FIFO register	32 bits	Read only
0x004	RX_DATA_FIFO	Receive packet-data FIFO register	32 bits	Read only
0x008	TX_LEN_FIFO	Transmit packet-length FIFO register	32 bits	Read/write
0x00C	TX_DATA_FIFO	Transmit packet-data FIFO register	32 bits	Read/write
0x010 – 0x0FC	Reserved			
Control and Status Registers				
0x100	VERID	Version ID register	32 bits	Read only
0x104	INTR_SRC	Interrupt source register	32 bits	Read only, clear on read
0x108	INTR_ENB	Interrupt enable register	32 bits	Read/write
0x10C	RX_STATUS	Receive status register	32 bits	Read only, clear on read
0x110	TX_STATUS	Transmit status register	32 bits	Read only, clear on read
0x114	RX_FRAMES_CNT	Receive frames counter register	32 bits	Read only

Table 5: LatticeMico32 Tri-Speed MAC Register Descriptions

Address [11:0] (Note: only [11:2] is decoded)	Register Name	Description	Size	Type
0x118	TX_FRAMES_CNT	Transmit frames counter register	32 bits	Read only
0x11C	RX_FIFO_TH	Receive FIFO threshold register	32 bits	Read/write
0x120	TX_FIFO_TH	Transmit FIFO threshold register	32 bits	Read/write
0x124	SYS_CTL	System control register	32 bits	Read/write
0x128	PAUSE_TMR	Pause timer register	32 bits	Read/write
0x12C – 0x1FC	Reserved			
Tri-Speed MAC Registers				
0x200	MAC_REGS_DATA	Tri-Speed MAC registers data register	32 bits	Read/write
0x204	MAC_REGS_ADDR_RW	Tri-Speed MAC registers address and read/write control register	32 bits	Read/write
0x208 – 0x2FC	Reserved			
0x00	MODE_BYTE0 MODE_BYTE1	Mode register	8 bits 8 bits	Read/write
0x02	TX_RX_CTL_BYTE0 TX_RX_CTL_BYTE1	Transmit and receive control register	8 bits 8 bits	Read/write
0x04	MAX_PKT_SIZE_BYTE0 MAX_PKT_SIZE_BYTE1	Maximum packet size register	8 bits 8 bits	Read/write
0x06	NULL_BYTE NULL_BYTE			
0x08	IPG_VAL_BYTE0 IPG_VAL_BYTE1	Inter-packet gap register	8 bits 8 bits	Read/write
0x0A	MAC_ADDR_0_BYTE0 MAC_ADDR_0_BYTE1	Tri-Speed MAC address register 0	8 bits 8 bits	Read/write
0x0C	MAC_ADDR_1_BYTE0 MAC_ADDR_1_BYTE1	Tri-Speed MAC address register 1	8 bits 8 bits	Read/write
0x0E	MAC_ADDR_2_BYTE0 MAC_ADDR_2_BYTE1	Tri-Speed MAC address register 2	8 bits 8 bits	Read/write
0x10	NULL_BYTE NULL_BYTE		8 bits 8 bits	Read/write

Table 5: LatticeMico32 Tri-Speed MAC Register Descriptions

Address [11:0] (Note: only [11:2] is decoded)	Register Name	Description	Size	Type
0x12	TX_RX_STS_BYTE0	Transmit and receive status	8 bits	Read only
	TX_RX_STS_BYTE1		8 bits	
0x14	GMII_MNG_CTL_BYTE0	GMII management interface control register	8 bits	Read/write
	GMII_MNG_CTL_BYTE1		8 bits	
0x16	GMII_MNG_DAT_BYTE0	GMII management data register	8 bits	Read/write
	GMII_MNG_DAT_BYTE1		8 bits	
0x18	NULL_BYTE			
	NULL_BYTE			
0x22	MLT_TAB_0_BYTE0	Multicast_table_0	8 bits	Read/write
	MLT_TAB_0_BYTE1		8 bits	
0x24	MLT_TAB_1_BYTE0	Multicast_table_1	8 bits	Read/write
	MLT_TAB_1_BYTE1		8 bits	
0x26	MLT_TAB_2_BYTE0	Multicast_table_2	8 bits	Read/write
	MLT_TAB_2_BYTE1		8 bits	
0x28	MLT_TAB_3_BYTE0	Multicast_table_3	8 bits	Read/write
	MLT_TAB_3_BYTE1		8 bits	
0x2A	MLT_TAB_4_BYTE0	Multicast_table_4	8 bits	Read/write
	MLT_TAB_4_BYTE1		8 bits	
0x2C	MLT_TAB_5_BYTE0	Multicast_table_5	8 bits	Read/write
	MLT_TAB_5_BYTE1		8 bits	
0x2E	MLT_TAB_6_BYTE0	Multicast_table_6	8 bits	Read/write
	MLT_TAB_6_BYTE1		8 bits	
0x30	MLT_TAB_7_BYTE0	Multicast_table_7	8 bits	Read/write
	MLT_TAB_7_BYTE1		8 bits	
0x32	VLAN_TAG_BYTE0	VLAN tag length/type	8 bits	Read only
	VLAN_TAG_BYTE1		8 bits	
0x34	PAUS_OP_BYTE0	Pause_opcode register	8 bits	Read/write
	PAUS_OP_BYTE1		8 bits	
Receive Statistics Counters				
0x300	RX_PKT_IGNR_CNT	Receive packet ignored counter register	32 bits	Read only, rollover
0x304	RX_LEN_CHK_ERR_CNT	Receive length check error counter	32 bits	Read only, rollover

Table 5: LatticeMico32 Tri-Speed MAC Register Descriptions

Address [11:0] (Note: only [11:2] is decoded)	Register Name	Description	Size	Type
0x308	RX_LNG_FRM_CNT	Receive long frames counter register	32 bits	Read only, rollover
0x30C	RX_SHRT_FRM_CNT	Receive short frames counter register	32 bits	Read only, rollover
0x310	RX_IPG_VIOL_CNT	Receive IPG violations counter register	32 bits	Read only, rollover
0x314	RX_CRC_ERR_CNT	Receive CRC errors counter register	32 bits	Read only, rollover
0x318	RX_OK_PKT_CNT	Receive OK packets counter register	32 bits	Read only, rollover
0x31C	RX_CTL_FRM_CNT	Receive control frame counter register	32 bits	Read only, rollover
0x320	RX_PAUSE_FRM_CNT	Receive pause frame counter register	32 bits	Read only, rollover
0x324	RX_MULTICAST_CNT	Receive multicast frame counter register	32 bits	Read only, rollover
0x328	RX_BRDCAST_CNT	Receive broadcast frame counter register	32 bits	Read only, rollover
0x32C	RX_VLAN_TAG_CNT	Receive VLAN tagged frame counter register	32 bits	Read only, rollover
0x330	RX_PRE_SHRINK_CNT	Receive preamble shrink counter register	32 bits	Read only, rollover
0x334	RX_DRIB_NIB_CNT	Receive dribble nibble counter register	32 bits	Read only, rollover
0x338	RX_UNSUP_OPCODE_CNT	Receive unsupported opcode counter register	32 bits	Read only, rollover
0x33C	RX_BYTE_CNT	Receive byte counter register	32 bits	Read only, rollover
0x340 – 0x3FC		Reserved		
Transmit Statistics Counters				
0x400	TX_UNICAST_CNT	Transmit unicast frame counter register	32 bits	Read only, rollover

Table 5: LatticeMico32 Tri-Speed MAC Register Descriptions

Address [11:0] (Note: only [11:2] is decoded)	Register Name	Description	Size	Type
0x404	TX_PAUSE_FRM_CNT	Transmit pause frame counter register	32 bits	Read only, rollover
0x408	TX_MULTICAST_CNT	Transmit multicast frame counter register	32 bits	Read only, rollover
0x40C	TX_BRDCAST_CNT	Transmit broadcast frame counter register	32 bits	Read only, rollover
0x410	TX_VLAN_TAG_CNT	Transmit VLAN tagged frame counter register	32 bits	Read only, rollover
0x414	TX_BAD_FCS_CNT	Transmit BAD FCS frame counter register	32 bits	Read only, rollover
0x418	TX_JUMBO_CNT	Transmit jumbo frame counter register	32 bits	Read only, rollover
0x41C	TX_BYTE_CNT	Transmit byte counter register	32 bits	Read only, rollover
0x420 – 0x4FC		Reserved		

Register Bit Definitions

Receive Packet-Length FIFO Register

Receive packet-length FIFO register (RX_LEN_FIFO) – 0x000, default value = undefined, read only

The receive packet-length FIFO register is at fixed address 0x000. It contains the length of the next packet in the receive data FIFO, in bytes. For example, a value of 0x59 read from this register indicates that there is an 89-byte packet in the RX_DATA_FIFO that must be read. This register is intended to be read only. After this register is read, RX_DATA_FIFO must be read.

31	30	29	28	27	26	25	24
23	22	21	20	19	18	17	16

15	14	13	12	11	10	9	8
Len_bit15	Len_bit14	Len_bit13	Len_bit12	Len_bit11	Len_bit10	Len_bit9	Len_bit8
7	6	5	4	3	2	1	0
Len_bit7	Len_bit6	Len_bit5	Len_bit4	Len_bit3	Len_bit2	Len_bit1	Len_bit0
Bits		Mnemonic		Description			
31:24		Not_used		Reserved			
23:16		Not_used		Reserved			
15:8		Len_byte1		High-order byte of length field			
7:0		Len_byte0		Low-order byte of length field			

Receive Packet-Data FIFO Register

Receive packet-data FIFO register (RX_DATA_FIFO) – 0x004, default value = undefined, read only

The receive packet-data FIFO register is at fixed address 0x004 and is intended to be read only. After reading the RX_LEN_FIFO register, 32-bit double words can be successively read from this register location.

31	30	29	28	27	26	25	24
Rx_data	Rx_data	Rx_data	Rx_data	Rx_data	Rx_data	Rx_data	Rx_data
Byte_3	Byte_3	Byte_3	Byte_3	Byte_3	Byte_3	Byte_3	Byte_3
Bit_7	Bit_6	Bit_5	Bit_4	Bit_3	Bit_2	Bit_1	Bit_0
23	22	21	20	19	18	17	16
Rx_data	Rx_data	Rx_data	Rx_data	Rx_data	Rx_data	Rx_data	Rx_data
Byte_2	Byte_2	Byte_2	Byte_2	Byte_2	Byte_2	Byte_2	Byte_2
Bit_7	Bit_6	Bit_5	Bit_4	Bit_3	Bit_2	Bit_1	Bit_0
15	14	13	12	11	10	9	8
Rx_data	Rx_data	Rx_data	Rx_data	Rx_data	Rx_data	Rx_data	Rx_data
Byte_1	Byte_1	Byte_1	Byte_1	Byte_1	Byte_1	Byte_1	Byte_1
Bit_7	Bit_6	Bit_5	Bit_4	Bit_3	Bit_2	Bit_1	Bit_0

Bits	Mnemonic	Description
31:24	Not_used	Reserved
23:16	Not_used	Reserved
15:8	Len_byte1	High-order byte of length field
7:0	Len_byte0	Low-order byte of length field

Transmit Packet-Data FIFO Register

Transmit packet-data FIFO register (RX_DATA_FIFO) – 0x00C, default value = N/A, write only

The transmit packet-data FIFO register is at fixed address 0x00C and is intended to be write only. After the TX_LEN_FIFO register is written to, the transmit FIFO extraction logic reads the length from the transmit length FIFO and then successively reads double words from this register location to send to the Tri-Speed MAC.

31	30	29	28	27	26	25	24
Tx_data	Tx_data	Tx_data	Tx_data	Tx_data	Tx_data	Tx_data	Tx_data
Byte_3	Byte_3	Byte_3	Byte_3	Byte_3	Byte_3	Byte_3	Byte_3
Bit_7	Bit_6	Bit_5	Bit_4	Bit_3	Bit_2	Bit_1	Bit_0

23	22	21	20	19	18	17	16
Tx_data	Tx_data	Tx_data	Tx_data	Tx_data	Tx_data	Tx_data	Tx_data
Byte_2	Byte_2	Byte_2	Byte_2	Byte_2	Byte_2	Byte_2	Byte_2
Bit_7	Bit_6	Bit_5	Bit_4	Bit_3	Bit_2	Bit_1	Bit_0

15	14	13	12	11	10	9	8
Tx_data	Tx_data	Tx_data	Tx_data	Tx_data	Tx_data	Tx_data	Tx_data
Byte_1	Byte_1	Byte_1	Byte_1	Byte_1	Byte_1	Byte_1	Byte_1
Bit_7	Bit_6	Bit_5	Bit_4	Bit_3	Bit_2	Bit_1	Bit_0

7	6	5	4	3	2	1	0
Tx_data	Tx_data	Tx_data	Tx_data	Tx_data	Tx_data	Tx_data	Tx_data
Byte_0	Byte_0	Byte_0	Byte_0	Byte_0	Byte_0	Byte_0	Byte_0
Bit_7	Bit_6	Bit_5	Bit_4	Bit_3	Bit_2	Bit_1	Bit_0

Bits	Mnemonic	Description
31:24	Tx_data_byte_3	Transmit data byte received fourth
23:16	Tx_data_byte_2	Transmit data byte received third
15:8	Tx_data_byte_1	Transmit data byte received second
7:0	Tx_data_byte_0	Transmit data byte received first

Version/Identification Register

Version/Identification register (VERID) – 0x010, default value = {0x0001, ID}, read only

The version/identification register is a read-only register that contains the version and identification fields. The upper 16 bits are a hard-coded Lattice Semiconductor version number. The identification is a 16-bit, user-defined parameter for the wrapper. You can set the wrapper identification in the `wb_wrapper.v` file before the `ctl_status_regs` instance.

31	30	29	28	27	26	25	24
Ver_bit15	Ver_bit14	Ver_bit13	Ver_bit12	Ver_bit11	Ver_bit10	Ver_bit9	Ver_bit8
23	22	21	20	19	18	17	16
Ver_bit7	Ver_bit6	Ver_bit5	Ver_bit4	Ver_bit3	Ver_bit2	Ver_bit1	Ver_bit0
15	14	13	12	11	10	9	8
ID_bit15	ID_bit14	ID_bit13	ID_bit12	ID_bit11	ID_bit10	ID_bit9	ID_bit8
7	6	5	4	3	2	1	0
ID_bit7	ID_bit6	ID_bit5	ID_bit4	ID_bit3	ID_bit2	ID_bit1	ID_bit0

Bits	Mnemonic	Description
31:24	Version_byte1	Hard-coded Lattice Semiconductor version number set to 0x00
23:16	Version_byte0	Hard-coded Lattice Semiconductor version number set to 0x01
15:8	ID_byte1	Identification bits that you set by using the ID_PARM parameter
7:0	ID_byte0	Identification bits that you set by using the ID_PARM parameter

Interrupt Source Register

Interrupt source register (INTR_SRC) – 0x014, default value = 0x0000_0000, read only, clear on read

The interrupt source register contains the receive and transmit interrupt sources. It is a read-only register, which is cleared on a read. The Tri-Speed MAC performs a logical AND on the interrupt bits from this register and the enable bits in the interrupt enable register. See “Interrupt Hierarchy” on page 36 for more details.

31	30	29	28	27	26	25	24
23	22	21	20	19	18	17	16
						RX_smry	TX_smry
15	14	13	12	11	10	9	8
			Rx_fifo_full	Rx_error	Rx_fifo_error	Rx_fifo_almost_full	Rx_pkt_rdy
7	6	5	4	3	2	1	0
			Tx_fifo_full	Tx_discfrm	Tx_fifo_almost_empty	Tx_fifo_almost_full	Tx_pkt_sent

Bits	Mnemonic	Description
31:24	Not_used	Reserved
23:18	Not_used	Reserved
17	Rx_smry	Receive interrupt source summary bit
16	Tx_smry	Transmit interrupt source summary bit
15:13	Not_used	Reserved
12	Rx_fifo_full	Receive FIFO is full (error condition)
11	Rx_error	Tri-Speed MAC detected receive packet error
10	Rx_fifo_error	Receive FIFO error condition
9	Rx_fifo_almost_full	Receive FIFO is almost full (based on threshold set)
8	Rx_pkt_rdy	Receive FIFO is ready (completely loaded in receive FIFO)

Bits	Mnemonic	Description
7:5	Not_used	Reserved
4	Tx_fifo_full	Transmit FIFO is full (error condition)
3	Tx_discrm	Tri-Speed MAC detected a transmit packet error
2	Tx_fifo_almost_empty	Transmit FIFO is almost empty (based on threshold set)
1	Tx_fifo_almost_full	Transmit FIFO is almost full (based on threshold set)
0	Tx_pkt_sent	Transmit packet sent to Tri-Speed MAC indication

Interrupt Enable Register

Interrupt enable register (INTR_ENB) – 0x018, default value = 0x0000_0000, read/write

The interrupt enable register is used in conjunction with the interrupt source register. The Tri-Speed MAC performs a logical AND on the interrupt bits from this register and the bits from the interrupt source register. See “Interrupt Hierarchy” on page 36 for more details. This register is read/write.

31	30	29	28	27	26	25	24
23	22	21	20	19	18	17	16
					Intr_enb	RX_smry_enb	TX_smry_enb
15	14	13	12	11	10	9	8
			Enb_12	Enb_11	Enb_10	Enb_9	Enb_8
7	6	5	4	3	2	1	0
			Enb_4	Enb_3	Enb_2	Enb_1	Enb_0

Bits	Mnemonic	Description
31:24	Not_used	Reserved
23:19	Not_used	Reserved
18	Intr_enb	Interrupt enable

Bits	Mnemonic	Description
17	Rx_smry_enb	Receive interrupt source summary bit enable
16	Tx_smry_enb	Transmit interrupt source summary bit enable
15:13	Not_used	Reserved
12	Rx_fifo_full	Interrupt enable
11	Rx_error	Interrupt enable
10	Rx_fifo_error	Interrupt enable
9	Rx_fifo_almost_full	Interrupt enable
8	Rx_pkt_rdy	Interrupt enable
7:5	Not_used	Reserved
4	Tx_fifo_full	Interrupt enable
3	Tx_discfrm	Interrupt enable
2	Tx_fifo_almost_empty	Interrupt enable
1	Tx_fifo_almost_full	Interrupt enable
0	Tx_pkt_sent	Interrupt enable

Receive Status Register

Receive status register (RX_STATUS) – 0x01C, default value = 0x0000_0000, read only, clear on read

The receive status register contains the receive status bits. It is a mirror of the receive portion of the interrupt source register and can be used for software polling and so forth without affecting the interrupt source register. This register is a read-only register, which is cleared on a read.

31	30	29	28	27	26	25	24
23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8

7	6	5	4	3	2	1	0
			Rx_fifo_full	Rx_error	Rx_fifo_error	Rx_fifo_almost_full	Rx_pkt_rdy

Bits	Mnemonic	Description
31:5	Not_used	Reserved
4	Rx_fifo_full	Receive FIFO is full (error condition)
3	Rx_error	Tri-Speed MAC detected receive packet error
2	Rx_fifo_error	Receive FIFO error condition
1	Rx_fifo_almost_full	Receive FIFO is almost full (based on threshold set)
0	Rx_pkt_rdy	Receive packet is ready (completely loaded in receive FIFO)

Transmit Status Register

Transmit status register (INTR_SRC) – 0x020, default value = 0x0000_0000, read only, clear on read

The transmit status register contains the transmit status bits. It is a mirror of the transmit portion of the interrupt source register and can be used for software polling and so forth without affecting the interrupt source register. This register is a read-only register, which is cleared on a read.

31	30	29	28	27	26	25	24
23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8
7	6	5	4	3	2	1	0
			Tx_fifo_full	Tx_discfrm	Tx_fifo_almost_empty	Tx_fifo_almost_full	Tx_pkt_rdy

Bits	Mnemonic	Description
31:5	Not_used	Reserved
4	Tx_fifo_full	Transmit FIFO is full (error condition)
3	Tx_discfrm	Tri-Speed MAC detected transmit packet error
2	Tx_fifo_almost_empty	Transmit FIFO is almost empty (based on threshold set)
1	Tx_fifo_almost_full	Transmit FIFO is almost full (based on threshold set)
0	Tx_pkt_sent	Transmit packet sent to Tri-Speed MAC indication

Receive Frames Count Register

Receive frames count register (RX_FRAMES_CNT) – 0x024, default value = 0x0000_0000, read only

The receive frames count register contains the number of frames currently in the receive data FIFO. The RX_FRM_CNT register is incremented by one each time the receive-length FIFO is written to and decremented by one each time the receive-length FIFO is read. This register is read only.

31	30	29	28	27	26	25	24
23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8
7	6	5	4	3	2	1	0
Rxfrm_cnt7	Rxfrm_cnt6	Rxfrm_cnt5	Rxfrm_cnt4	Rxfrm_cnt3	Rxfrm_cnt2	Rxfrm_cnt1	Rxfrm_cnt0

Bits	Mnemonic	Description
31:24	Not_used	Reserved
23:16	Not_used	Reserved

Bits	Mnemonic	Description
15:8	Not_used	Reserved
7:0	Rxfrm_cnt	Receive frame counter bits 7:0 - count valid from 0 to 255

Transmit Frames Count Register

Transmit frames count register (TX_FRAMES_CNT) – 0x028, default value = 0x0000_0000, read only

The transmit frames count register contains the number of frames currently in the transmit data FIFO. The TX_FRM_CNT register is incremented by one each time the transmit length FIFO is written to and decremented by one each time a transmit packet is transmitted by the Tri-Speed MAC (transmit-length FIFO is read and packet sent). This register is read only.

31	30	29	28	27	26	25	24
23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8
7	6	5	4	3	2	1	0
Txfrm_cnt7	Txfrm_cnt6	Txfrm_cnt5	Txfrm_cnt4	Txfrm_cnt3	Txfrm_cnt2	Txfrm_cnt1	Txfrm_cnt0

Bits	Mnemonic	Description
31:24	Not_used	Reserved
23:16	Not_used	Reserved
15:8	Not_used	Reserved
7:0	Txfrm_cnt	Transmit frame counter bits 7:0 - count valid from 0 to 255

Receive FIFO Threshold Register

Receive FIFO threshold register (RX_FIFO_TH) – 0x11C, default value = 0x0000_0000, read/write

The receive FIFO threshold register is a read/write register used to set the almost empty and almost full threshold watermarks for the receive-data FIFO. The receive-data FIFO is composed of four cascaded 512x32 FIFOs to effectively build a 2048x32 FIFO (8192 bytes). The almost full and almost empty thresholds, therefore, are set on the basis of 512 locations for the top and bottom of the four cascaded FIFOs. A 9-bit value represents these watermarks.

31	30	29	28	27	26	25	24
AE_TH_b8							
23	22	21	20	19	18	17	16
AE_TH_b7	AE_TH_b6	AE_TH_b5	AE_TH_b4	AE_TH_b3	AE_TH_b2	AE_TH_b1	AE_TH_b0
15	14	13	12	11	10	9	8
AF_TH_b8							
7	6	5	4	3	2	1	0
AF_TH_b7	AF_TH_b6	AF_TH_b5	AF_TH_b4	AF_TH_b3	AF_TH_b2	AF_TH_b1	AF_TH_b0
Bits	Mnemonic			Description			
31:25	Not_used			Reserved			
24:16	AE[8:0]			RX_DATA FIFO almost empty watermark bits			
15:9	Not_used			Reserved			
8:0	AF[8:0]			RX_DATA FIFO almost full watermark bits			

Transmit FIFO Threshold Register

Transmit FIFO threshold register (TX_FIFO_TH) – 0x120, default value = 0x0000_0000, read/write

The transmit FIFO threshold register is a read/write register used to set the almost empty and almost full threshold watermarks for the transmit-data FIFO. The transmit-data FIFO is composed of four cascaded 512x32 FIFOs to effectively build a 2048x32 FIFO (8192 bytes). The almost full and almost empty thresholds, therefore, are set on the basis of 512 locations for the top

and bottom of the four cascaded FIFOs. A 9-bit value represents these watermarks.

31	30	29	28	27	26	25	24
AE_TH_b8							
23	22	21	20	19	18	17	16
AE_TH_b7	AE_TH_b6	AE_TH_b5	AE_TH_b4	AE_TH_b3	AE_TH_b2	AE_TH_b1	AE_TH_b0
15	14	13	12	11	10	9	8
AF_TH_b8							
7	6	5	4	3	2	1	0
AF_TH_b7	AF_TH_b6	AF_TH_b5	AF_TH_b4	AF_TH_b3	AF_TH_b2	AF_TH_b1	AF_TH_b0
Bits	Mnemonic			Description			
31:25	Not_used			Reserved			
24:16	AE[8:0]			TX_DATA FIFO almost empty watermark bits			
15:9	Not_used			Reserved			
8:0	AF[8:0]			TX_DATA FIFO almost full watermark bits			

System Control Register

System control register (SYS_CTL) – 0x124, default value = 0x0000_0000, read/write

The system control register is a read/write register used to control the Tri-Speed MAC and receive and transmit FIFOs for system-related tasks.

31	30	29	28	27	26	25	24
23	22	21	20	19	18	17	16

15	14	13	12	11	10	9	8
7	6	5	4	3	2	1	0
			Tx_fifo_flush	Rx_fifo_flush	Tx_sndpausreq	TX_fifoctrl	Ignore_next_pkt

Bits	Mnemonic	Description
31:5	Not_used	Reserved
4	Tx_fifo_flush	Flushes the transmit FIFO
3	Rx_fifo_flush	Flushes the receive FIFO
2	Tx_sndpausreq	When the pause frame request is asserted, the Tri-Speed MAC transmits a pause frame. This is also the qualifying signal for the tx_sndpausetim bus.
1	Tx_fifoctrl	The FIFO control frame signal indicates whether the current frame in the transmit FIFO is a control frame or a data frame. The following values apply: ◆ 1 = Control frame ◆ 0 = Normal frame
0	Ignore_next_pkt	The ignore next packet bit should be asserted by the host to prevent a receive FIFO full condition. The receive Tri-Speed MAC continues dropping packets as long as this bit is asserted. This bit can be set asynchronously to the Tri-Speed MAC operation.

Pause Timer Register

Pause timer register (PAUSE_TMR) – 0x128, default value = 0x0000_0000, read/write

The pause timer register is a read/write register used to set the pause time value that should be sent in the pause frame sent by the Tri-Speed MAC. Two different methods are used for transmitting a pause frame:

- ◆ The application layer can form a pause frame and submit it for transmission through the FIFO.
- ◆ The application layer signals the transmit Tri-Speed MAC directly to transmit a pause frame by asserting the tx_sndpausreq bit in the sys_ctl register. In this case, the transmit Tri-Speed MAC completes transmission

of the current packet and transmits a pause frame with the pause time value supplied through the value in this pause_tmr register.

31	30	29	28	27	26	25	24
23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8
PT_bit15	PT_bit14	PT_bit13	PT_bit12	PT_bit11	PT_bit10	PT_bit9	PT_bit8
7	6	5	4	3	2	1	0
PT_bit7	PT_bit6	PT_bit5	PT_bit4	PT_bit3	PT_bit2	PT_bit1	PT_bit0
Bits		Mnemonic			Description		
31:16		Not_used			Reserved		
15:10		PT[15:0]			Pause timer value		

Tri-Speed MAC Registers Data Register

Tri-Speed MAC registers data register (MAC_REGS_DATA) – 0x200, default value = 0x0000_0000, read/write

The Tri-Speed MAC registers data register is a read/write register that contains the data read back or the data to write to at the address specified in MAC_REG_ADDR.

31	30	29	28	27	26	25	24
23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8
BYTE1_7	BYTE1_6	BYTE1_5	BYTE1_4	BYTE1_3	BYTE1_2	BYTE1_1	BYTE1_0

7	6	5	4	3	2	1	0
BYTE0_7	BYTE0_6	BYTE0_5	BYTE0_4	BYTE0_3	BYTE0_2	BYTE0_1	BYTE0_0
Bits		Mnemonic		Description			
31:16		Not_used					
15:8		Data_high_byte		Read from or write to high-byte data Tri-Speed MAC registers specified by the address in MAC_REG_ADDR			
7:0		Data_low_byte		Read from or write to low-byte data Tri-Speed MAC registers specified by the address in MAC_REG_ADDR			

Tri-Speed MAC Registers Address and Read/Write Control Register

Tri-Speed MAC registers address and read/write control register (MAC_REGS_ADDR_RW) – 0x204, default value = 0x0000_0000, read/write

The Tri-Speed MAC registers address is a read/write register that contains the read or write address of the Tri-Speed MAC register to be accessed and a read/write control bit that indicates a read (low) or write (high) operation.

Note

The address written to this register is on word boundaries.

31	30	29	28	27	26	25	24
RW							
23	22	21	20	19	18	17	16
15	14	13	12	11	10	9	8
7	6	5	4	3	2	1	0
Add_7	Add_6	Add_b5	Add_4	Add_3	Add_2	Add_1	Add_0

Bits	Mnemonic	Description
31	Read/write bit	Control bit that indicates a read (low) or write (high) operation.
30:8	Unused	
7:0	Address	8-bit Tri-Speed MAC address

Timing Diagrams

Figure 4 shows the Tri-Speed MAC reading an 89-byte receive packet.

Figure 4: Tri-Speed MAC Reading an 89-Byte Rx Packet

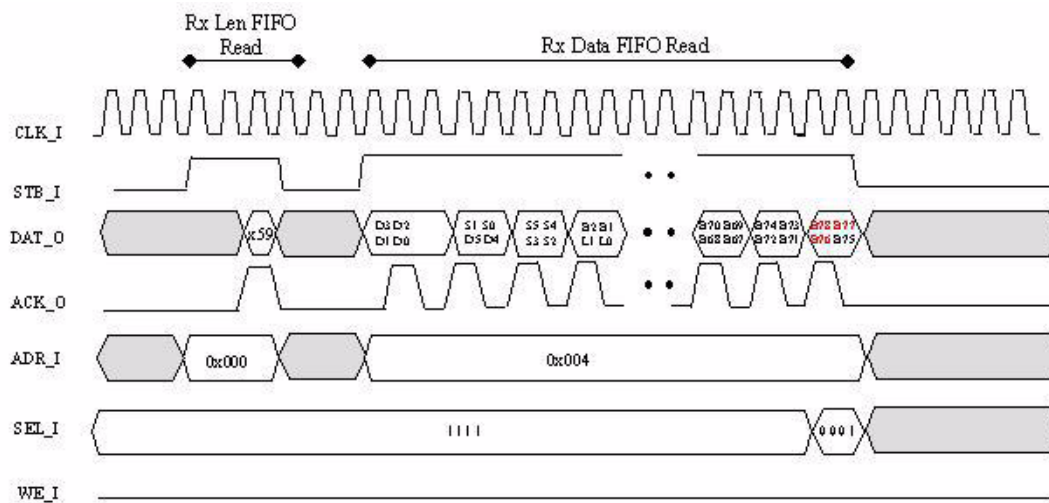


Figure 5 shows the Tri-Speed MAC writing an 80-byte transmit packet.

Figure 5: Tri-Speed MAC Writing an 80-Byte Tx Packet

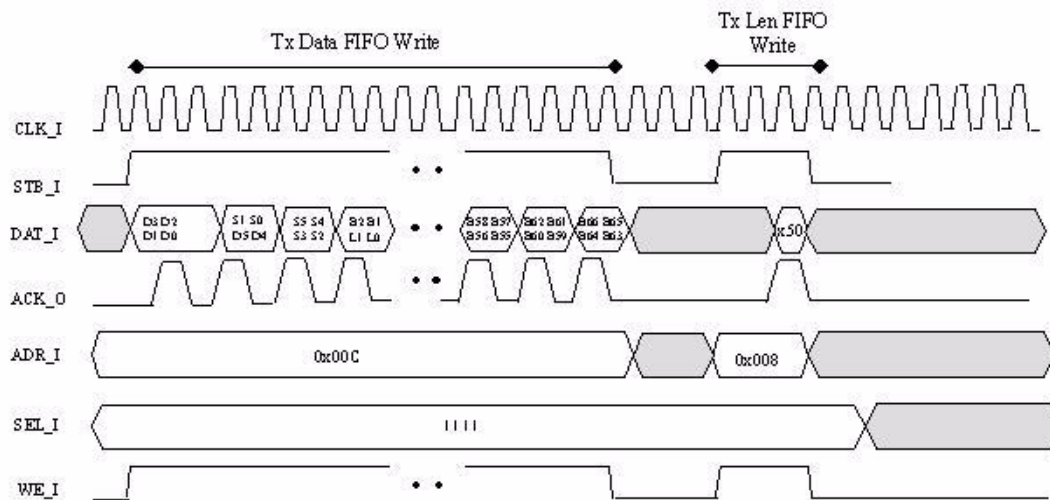
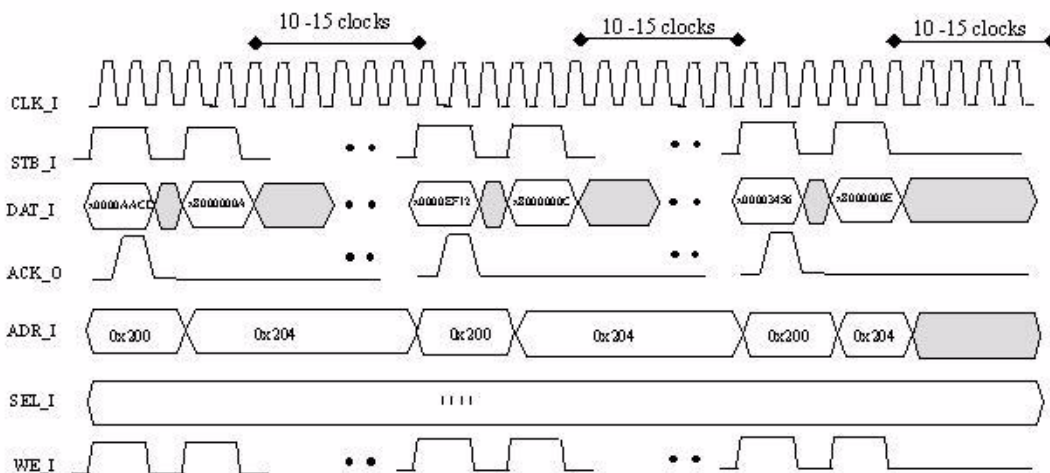


Figure 6 shows the Tri-Speed MAC writing a MAC address to AA-CD-EF_12-34-56.

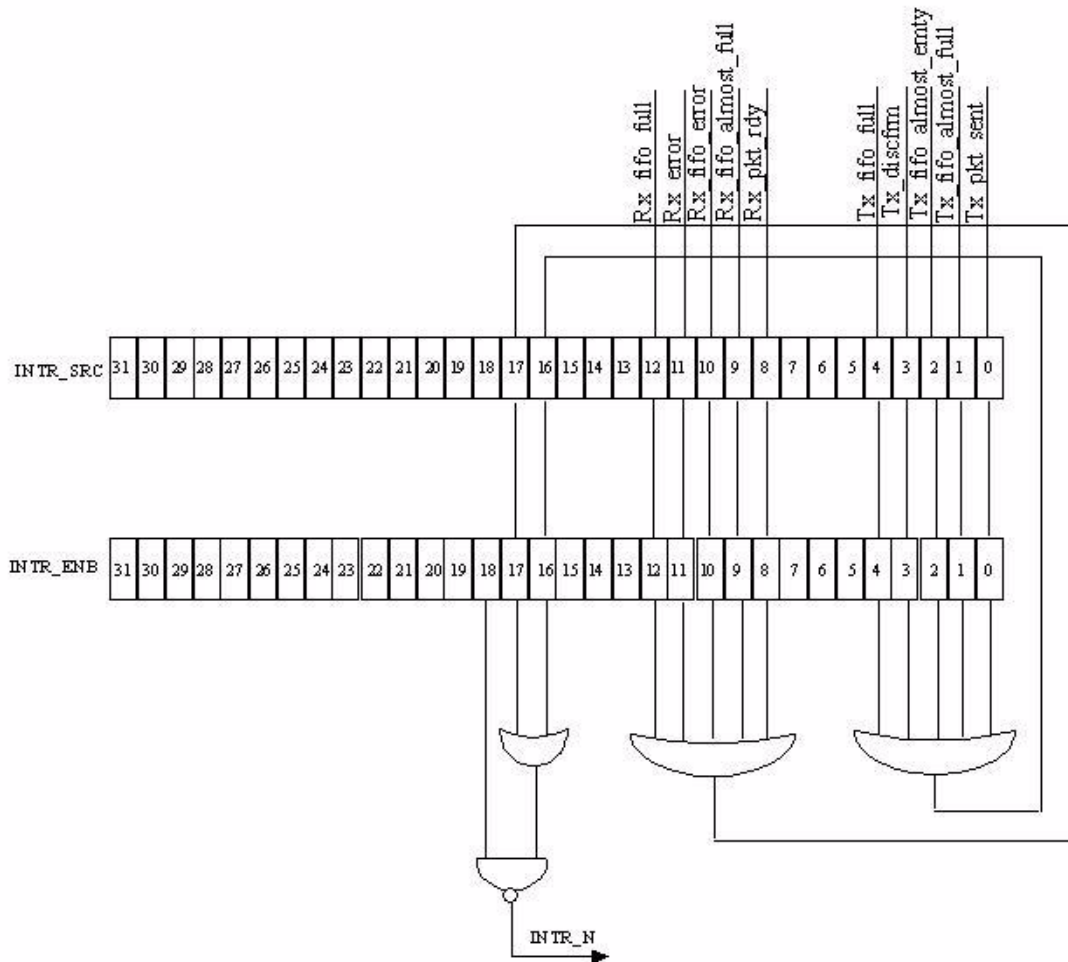
Figure 6: Tri-Speed MAC Writing a MAC Address



Interrupt Hierarchy

Figure 7 shows the interrupt hierarchy used.

Figure 7: Tri-Speed MAC Interrupt Hierarchy



Additional Information

Following is some additional information about the LatticeMico32 Tri-Speed MAC:

- ◆ It is assumed that any data structures used by the TCP/IP layer, like linked lists and their maintenance, is handled by the LatticeMico32 software.
- ◆ It is assumed that the additional processor clock cycles involved in reading and writing control registers through the register control path and any additional software processing does not adversely impact the throughput requirements.

- ◆ The depth of the FIFO buffers is 8 kilobytes.
- ◆ The full set of IEEE 802.3 statistics counters and registers is not supported. A subset of the statistics counters is implemented. You can easily remove this module with the Include Statistics Regs option in the graphical user interface.
- ◆ $CLK_I = 2 \times (MAC_byte \text{ clock})$, at a minimum. That is, there is at least a 2x frequency differential between the Tri-Speed MAC byte clocks, such as rxmac_clk and txmac_clk, and the WISHBONE clock, CLK_I, for reliable operation. For example, if the Tri-Speed MAC is operating in 100-MB mode with rxmac_clk and txmac_clk equal to 12.5 MHz, CLK_I must be equal or greater than 25 MHz.
- ◆ This core, in conjunction with LatticeMico32, is suitable for fast Ethernet applications (100 megabits per second) only. One-gigabit applications would require you to modify the existing design.

Reference Information

The following documents provide more information on the Tri-Speed MAC:

- ◆ *IPUG51 - Tri-Speed Ethernet Media Access Controller User's Guide*
- ◆ *IEEE 802.3 Ethernet Specifications for 10/100/1G*

You can also obtain more information at the following Web site:

<http://www.latticesemi.com/products/intellectualproperty/ipcores/mico32/>

